

PROBLEM STATEMENT

Time-3 hours

Create a chatbot that:

- Maintains conversation history
- Gives context-aware responses
- Avoids repeating answers

Example:

User: “Tell me about Python”

User: “What are its advantages?” → Bot should understand “its” = Python

Requirements

- Use LLM API
- Maintain:
- Chat history (in-memory or simple DB)
- Implement:
- Context window handling
- Token optimization (truncate/summarize history)

Prompt Engineering Task

- Create a context-aware prompt template
- Add:
- Instruction to avoid hallucination
- Instruction to ask clarification if unsure

Demo Requirement

- Show a multi-turn conversation

Demonstrate:

- Context retention
- Follow-up understanding.